

SRIMATHI MAHALINGAM

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EDUCATION

University of Massachusetts, Amherst

08/23 – 05/25

Master of Science in Computer Science (Specialization in Security Systems)

CGPA: 3.93/4.0

- Coursework: Secured Distributed Systems, Machine Learning, Systems for Deep Learning, Software Engineering, Advanced Natural Language Processing, Algorithms for Data Science, Introduction to Computer and Network Security

College of Engineering, Guindy, Anna University

08/17 – 05/21

Bachelor of Engineering in Computer Science Engineering

CGPA : 8.5/10.0

- Coursework: Data Structures, Compiler Design, Embedded Systems, Algorithms, Object Oriented Programming, Database Systems, Digital Principles and Design, Computer Architecture, Operating Systems, Digital Signal Processing

EXPERIENCE

Micron Technology

Software Developer II

08/22 – 03/23

- Led the NAND Open/Short Test programs in C++, cutting testing time by 40% with agile methodologies.
- Key contributor to Aries3 mNAND test script development, addressing architectural challenges and Micron's proprietary controller, demonstrating strong cross-functional collaboration and interpersonal skills.

Software Developer

06/21- 07/22

- Contributed to the development, debugging, and maintenance of C++ test programs for the Micron Test Firmware team, including API implementation and code reviews.
- Collaborated with various engineering and production groups to ensure test program, infrastructure and system readiness for new product introduction as well as high volume manufacturing.

Zoho Corporation

Software Intern

06/20 – 07/20

- Designed a language translation product utilizing HTML, CSS, JavaScript, jQuery, and Python Flask, resulting in a 25% increase in user adoption rates.
- Authored detailed documentation for the language translation feature, reducing onboarding time for new team members by 30%.
- Demonstrated full stack development skills by working on both frontend and backend components.

PROJECTS

Understanding Forward-Mode Auto Differentiation | CUDA, Python, PyTorch, NLP |

- Investigated the GPU memory usage for large model training using PyTorch and JAX capabilities with CUDA.
- Achieved a 15% increase in computational efficiency and reduced peak memory usage from 12.2 GB to 10.4 GB.
- Conducted comprehensive memory profiling and performance analysis leading to a 20% improvement in model optimization.

Emotion-Cause Pair Extraction (ECPE) | Python, NLP, Transformers, LLM |

- Developed advanced NLP models using BERT and OPT 350M for ECPE, achieving 73% accuracy in emotion detection.
- Implemented an end-to-end BiLSTM network, achieving 65% accuracy in extracting emotions their causes.
- Partnered with a team of 5 members to design experiments, enhancing communication skills.

Patient Tracker System | MySQL, ReactJS, Flask |

- Designed a web-based Patient Tracker System that streamlined healthcare operations, with 75% of user satisfaction survey respondents reporting improved ease of use and navigation.
- Contributed to all components of software development life cycle in compliance with best coding practices
- Employed MVC Architecture and performed thorough unit testing to assist in troubleshooting.

PUBLICATIONS

- Srimathi M, et al. “**Phishing URL Detection and Vulnerability Assessment of Web Applications Using IVS Attributes with Explainable Artificial Intelligence (XAI)**” ICT Analysis and Applications. Springer Series, Singapore, 2022. 

SKILLS

- **Programming Languages:** C#, C++, Python, Java, SQL, JavaScript/TypeScript
- **Web Technologies:** HTML/CSS, PHP, Django, Node.js, ReactJS, Android
- **ML/AI:** Pandas, NumPy, TensorFlow, Anaconda, Keras
- **Operating Systems:** Windows, Linux, Ubuntu, VMware
- **Tools and Frameworks:** Visual Studio, Version Control (Git), Bitbucket, JIRA, confluence, Unity, Tableau